Luqman **Bachtiar**

AI/ML Solutions · Generative & Agentic AI · Data & AI Leadership

Auckland, New Zealand

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Summary

Data & AI leader who takes AI systems from architecture to production and stays hands-on at every layer in between. 8+ years across data science, machine learning, and data engineering, now building production Generative and agentic AI.

Leads the Asia-Pacific data engineering function at Ever.Ag Data Labs, the AI arm of Ever.Ag (an agtech company bringing predictive intelligence to the global food and agriculture supply chain), owning customer delivery across New Zealand and Australia and growing the team. Built the firm's AI-assisted development capability from the ground up, cut design-to-deployment from weeks to days, and wrote the guardrails that let the rest of the team move that fast safely. Works hands-on in Python, LLM application development, fine-tuning and evaluation, MLOps, and cloud architecture on Azure, AWS, and GCP, turning ambiguous business problems into direction and measurable value for C-suite and operational stakeholders.

Skills and Attributes

Generative AI & LLMs	LLM application development (Claude, GPT), prompt engineering, and model selection, fine-tuning & evaluation.
Agentic AI	Tool-using, multi-step agent systems built directly against model APIs; AI-assisted development with Claude Code and Codex.
AI Governance & Responsible AI	Usage-guardrails frameworks, safe and compliant deployment, and AI usage policy for enterprise rollout.
AI Enablement & Leadership	Lifting AI fluency across technical and commercial teams; team leadership, mentoring, and internal skills curricula.
Business & Strategy	Spotting automation & augmentation opportunities, translating business needs into solution architecture, and validating measurable value.
Machine Learning	scikit-learn, TensorFlow/Keras, SparkML, Optuna hyper-parameter optimisation; forecasting, segmentation, and mathematical modelling.
MLOps & LLMOps	Productionising and monitoring ML/LLM systems across Azure & AWS; model evaluation and continuous improvement.
Programming & Frameworks	Python, SQL, PySpark, Bash, FastAPI, React, Git.
Data Engineering	Data pipeline architecture, data modelling, ETL, orchestration (Prefect), and API & SFTP integration.
Cloud & Containerisation	Microsoft Azure (MLOps, Synapse, Cognitive Services, DevOps), AWS (ECS Fargate, RDS, CloudFront, ALB), Google Cloud Platform; Docker.
Database Management	MySQL, Microsoft SQL Server, Teradata, Oracle Database, Google BigQuery, Databricks, SQLite.
Visualisation & Reporting	Tableau, PowerBI, Apache Superset, Excel.
Ways of Working	Agile, Software Development Life Cycle, Atlassian suite.

Selected Projects

Founder & Full-Stack Engineer · case study & demo

Fairway Plan *Launched; since retired*

- Plans optimised multi-day golf trips across New Zealand, choosing which courses to play and when, balancing course quality, budget, travel time, weather, and daylight with a mixed-integer-programming engine. Designed, built, and launched solo through an LLM-assisted workflow.
- Ran as a containerised microservices stack (Next.js, FastAPI, PostgreSQL, Redis, nginx) with a dedicated HiGHS mixed-integer-programming service driving the optimisation.
- Backed by automated data-enrichment and weather pipelines (370+ courses with multi-year history, caching, scheduled refresh) and a full Playwright end-to-end suite, deployed via Docker Compose to cloud infrastructure over HTTPS.

Founder & Full-Stack Engineer · case study & demo

Beanmarkd *Launched; since retired*

- A two-sided social review app for specialty coffee, an "Untappd for coffee beans." Drinkers log and rate their brews; roasters post fresh roast drops; bean freshness (days off roast) is the number that matters. Designed, built, and launched solo and LLM-assisted.
- React, FastAPI, and MySQL stack with JWT authentication, role-based and roaster-scoped permissions, parameterised SQL, versioned database migrations, and a layered test pyramid of 175+ automated tests across API, component, and end-to-end levels.

Creator & Sole Engineer · lrbachtiar.github.io/ferry

Ferry *Open source*

- A CLI that seeds test environments with production-shaped EHR data. Walks foreign keys to keep whole patient records intact, then masks the PHI deterministically, with zero dangling references left behind.
- Exports FHIR R4 and HL7 v2.5.1; built on DuckDB, Python, and Synthea. Draws directly on my healthcare-interoperability background at Orion Health.

Experience

Ever.Ag Data Labs *Auckland, New Zealand*

Data Engineering Manager *March 2026 - Present*

- Lead the Asia-Pacific data engineering team across New Zealand, Australia, and the Philippines; own NZ and AU customer delivery and grow the team.
- Built the AI-assisted development capability from the ground up, cutting design-to-deployment from weeks to days across the project portfolio.
- Embedded LLM-based tooling across the delivery lifecycle: product development, production deployment, large-scale migrations, and customer-facing sales engagements.
- Wrote the guardrails that let the team move fast safely: a training programme, usage-guardrails framework, and fortnightly AI working group spanning engineering, data science, and sales.
- Shipped a manufacturing optimiser for a meat processor that aligns production to seasonal demand within real factory and capacity constraints, using a linear-programming solver with LLM-written commentary and bulk SFTP. Live and customer-facing.
- Prototyped a full-stack supply-chain platform (demand planning, supply optimisation, SFTP ingestion) on React, FastAPI, MySQL, and Prefect, backed by a production-grade AWS reference architecture (ECS Fargate, RDS, CloudFront, ALB).
- Cut sales-prototype build time from three weeks to four hours, accelerating new-customer acquisition.

Architecture & Cloud: AWS (ECS Fargate, RDS, CloudFront, ALB), API design, SFTP integration
AI Tools: LLM application development, Claude Code, Codex
Programming Languages: Python, SQL, PySpark, Bash
Data: MySQL, data modelling, Prefect
Delivery: Git, Playwright, Azure DevOps, Atlassian suite (Jira, Confluence)

Senior Data Engineer - Client-Facing Data Architect *August 2023 - February 2026*

- Deployed and integrated scAIcloud™, a data science platform spanning data ingestion, optimisation, and front-end configuration, for clients across the milk and meat sectors.
- Delivered livestock forecasting, planning, and sales forecasting solutions for meat producers.
- Translated complex data and mathematical problems into clear decisions for everyone from C-suite executives to day-to-day end users.

Toolkit: Same as above

Parrot Analytics

Auckland, New Zealand

Senior Data Scientist

July 2022 - July 2023

- Core member of the Data Science team at Parrot Analytics, which measures global supply and demand for entertainment content across film, television, gaming, and talent. Delivered R&D projects and client engagements end to end.
- Built pre-release demand forecasts for movies and TV shows.
- Linked movie Print & Advertising (P&A) spend to global title demand through correlation analysis.
- Predicted post-release movie demand (a proxy for box-office performance) from genomic metadata alone: genre, director, cast, and production company.
- Modelled consumer segmentation and churn for paid online streaming platforms.

Programming Languages: Python, SQL, PySpark, Bash
Databases and Tools: MySQL
Big Data Tools: Apache Spark
Development Tools: Git, Amazon Web Services
Other Tools: Atlassian suite (Jira, Confluence)

TVNZ - Television New Zealand

Auckland, New Zealand

Data Scientist

November 2020 - July 2022

- Owned the Advanced Analytics roadmap, capability, and technology stack for the News and Video-on-Demand (VOD) products, from user-event tagging design through to productionising machine learning models on Azure (MLOps).
- Ran VOD experimentation through structured A/B testing, and drove product insight from Google Analytics user-tracking and broadcast metrics (Youbora).
- Mentored junior analysts and founded an internal data-professionals community to lift data literacy across the organisation.
- Set up an academic partnership with the University of Auckland, supervising Master of Data Science students on industry-placement thesis projects.
- Recruited Product Analysts and screened Data Engineering candidates.
- Contributed to enterprise data governance through TVNZ's Data Governance Forum, and managed vendors via RACI and RFQ processes.

Programming Languages: Python, SQL, PySpark, Bash
Databases and Tools: Google BigQuery, Microsoft SQL Server
Big Data Tools: Apache Spark, Databricks, Azure Synapse
Machine Learning Tools: scikit-learn, SparkML, Amazon Personalize, Azure Cognitive Services (Text Analytics and Computer Vision), MLflow
Visualisation Tools: Tableau, PowerBI, Excel
Development Tools: Git, Amazon Web Services, Google Cloud Platform, Microsoft Azure
Other Tools: Google Analytics, Segment.io, Split.io, Agile Ways of Working, Atlassian suite (Jira, Confluence)

ANZ - Australia and New Zealand Banking Group Limited

Melbourne, Australia

Data Scientist

September 2019 - November 2020

- Built and rolled out the Data Chapter Syllabus, a role-based skills matrix and in-depth learning documentation, to upskill Campaign Analysts, Data Analysts, and Data Scientists.
- Delivered marketing analytics across campaign post-implementation reviews, trigger analysis, and predictive modelling.
- Delivered business analytics across customer acquisition, early engagement, utilisation, retention, and win-back.
- Automated an ETL pipeline managing high-net-worth customer data assets.
- Built a customer-acquisition model that generated data-driven high-net-worth leads by segmenting and modelling financial transaction data.

Programming Languages: Python, SQL, Bash, SAS
Databases and Tools: Teradata, Oracle Database, Microsoft SQL Server
Big Data Tools: Apache Spark, Apache Sqoop
Machine Learning Tools: scikit-learn, Spark MLlib
Visualisation Tools: Tableau, Excel
Development Tools: Git
Other Tools: Agile Ways of Working, Atlassian suite (Jira, Confluence)

Fonterra Co-operative Group Limited *Auckland, New Zealand*

Data Scientist *June 2018 - September 2019*

- Scrum Master for the Centre of Excellence Data Science team, driving agile delivery and ways of working.
- Led workshops and regular technical presentations to upskill data professionals across the business.
- Produced advanced-analytics insight from large-scale trade, manufacturing, and sales datasets.
- Designed, built, and deployed a big-data pipeline on AWS cloud infrastructure.
- Built descriptive and predictive models, surfacing outputs through Tableau dashboards for Sales Managers.
- Translated complex mathematical analysis into business decisions for stakeholders.
- Delivered projects valued at $40M (Global Sales and Account Opportunity Pipeline Dashboard), $14M (Modelling Product Sensory Quality Failure), $4M (Production Line Quality Simulation), and $3.5M (Vehicle Life Extension Model).
- Built Coffee Roulette, an employee socialising application.

Programming Languages: Python, SQL, R, Bash
Databases and Tools: Microsoft SQL Server, SQL Server Integration Services, SQLite
Big Data Tools: Apache Spark, Cloudera Data Science Workbench
Machine Learning Tools: Keras (TensorFlow backend)
Visualisation Tools: Tableau, Excel
Development Tools: Git, AWS Cloud Computing Services
Other Tools: Agile development methodology, Atlassian suite (Jira, Confluence)

Earlier Career

2018	**Health Integration Analyst: data quality and BI systems**, *St John New Zealand*	*Auckland, New Zealand*
2015 - 2018	**Solution Engineer: AWS deployment; HL7, CDA and FHIR ingestion**, *Orion Health*	*Auckland, New Zealand*
2015	**Implementation Consultant: HealthTerm integration**, *Orion Health*	*Auckland, New Zealand*
2010 - 2013	**Course Tutor and Marker: MATLAB and R**, *The University of Auckland*	*Auckland, New Zealand*
2010 - 2013	**Research Assistant: biosensor signal processing**, *Plant & Food Research; Auckland Bioengineering Institute*	*Auckland, New Zealand*
2009 - 2010	**Student Research Assistant: computer vision**, *UTeM, Malaysia*	*Melaka, Malaysia*

Education

The University of Auckland *Auckland, New Zealand*

Doctoral Research (PhD, not completed) *January 2013 - April 2019*

- Applied machine learning and neural networks to olfactory biosensor design, with resulting work published in peer-reviewed journals and IEEE conferences.
- Doctoral Research Scholarship from New Zealand's New Economy Research Fund.

The University of Auckland *Auckland, New Zealand*

Master of Engineering in Engineering Science *July 2011 - May 2013*

- First Class Honours
- Master's Research Scholarship from New Zealand's New Economy Research Fund

The University of Tokushima *Tokushima, Japan*

Summer School in Information Science and Intelligent Systems *August 2012*

- Summer School Exchange Scholarship

The University of Auckland *Auckland, New Zealand*

Bachelor of Engineering in Biomedical Engineering *February 2008 - September 2011*

- Second Class Honours
- Kickstarter Merit Scholarship

Publications

Journal Papers

Using Multilayer Perceptron Computation to Discover Ideal Insect Olfactory Receptor Combinations in the Mosquito and Fruit Fly for an Efficient Electronic Nose — *January 2015*

Luqman R. Bachtiar, Charles P. Unsworth and Richard D. Newcomb

Neural Computation, The MIT Press Journals

Multilayer Perceptron Classification of Unknown Volatile Chemicals from the Firing Rates of Insect Olfactory Sensory Neurons and Its Application to Biosensor Design — *January 2013*

Luqman R. Bachtiar, Charles P. Unsworth, Richard D. Newcomb and Edmund J. Crampin

Neural Computation, The MIT Press Journals

Conference Papers

Six papers at the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (2011–2016), applying multi-layer perceptrons and neural networks to olfactory biosensor design, plus a paper at the 2011 Joint Symposium on Neural Computation.

Awards

September 2013 — **Engineering Postgraduate Poster Competition 2013 People's Choice Award** — *Auckland, New Zealand*

Engineering Postgraduate Society

Awarded the People's Choice Award for the poster "Identifying Odorants Using Fly-Mosquito Olfactory Sensors and Multi-Layer Perceptron Classification Techniques for an Olfactory Biosensor Device" at the 2013 Engineering Postgraduate Poster Competition.

November 2010 — **University of Auckland Summer Studentship Scholarship** — *Auckland, New Zealand*

The University of Auckland

A Summer Research Scholarship for undergraduate students to undertake a research project over the summer.

November 2009 — **Universiti Teknikal Malaysia Melaka Summer Studentship Scholarship** — *Melaka, Malaysia*

Universiti Teknikal Malaysia Melaka

A Summer Research Scholarship for undergraduate students to undertake a research project over the summer.